Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Definitions

The questions in this practice assignment relate to the nonlinear spring-mass-damper system introduced in this lesson. For this system, assume the following definitions.

- $m = 50.$
- $k_1 = 4.$
- $k_2 = 60.$
- $d = 0.25.$
- $b = 2.$
- $\Delta t = 0.1.$
- $\hat{x}_k^- = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$
- $\hat{x}_k^+ = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$

1. Compute $\hat{A}_k$ for these definitions. What is the (2,1) entry of this matrix? (That is, what is the entry in the second row and first column?) Enter your answer with three digits to the right of the decimal point. — 1 / 1 point

-0.008

Correct
Yes, that is correct.

2. Compute $\hat{B}_k$ for these definitions. What is the (2,1) entry of this matrix? (That is, what is the entry in the second row and the first column?) Enter your answer with three digits to the right of the decimal point. — 1 / 1 point

0.002

Correct
Yes, that is correct.

3. Compute $\hat{C}_k$ for these definitions. What is the (1,1) entry of this matrix? (That is, what is the entry in the first row and first column?) Enter your answer with one digit to the right of the decimal point. — 1 / 1 point

4

Correct
Yes, that is correct.